

Using Computer Vision and Temporal Modeling to Infer Health from Chewing Behavior

Berkeley Master of Analytics

A video-based analytics system for chewing-health assessment and exploratory food-texture classification

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Introduction

Chewing behavior is a daily but underused signal for health monitoring. Patterns such as fast eating, low chew count, short bite duration, and one-sided chewing may reflect metabolic, behavioral, or oral-health risks. However, chewing assessment is usually manual, subjective, or device-dependent.

Thus, we aim to build a camera-based analytics pipeline that converts eating videos into structured chewing data, interpretable health scores, and personalized recommendations.

Methods

We developed a two-branch pipeline from the same video-derived chewing signals after extracting mouth and jaw landmarks.

Branch 1: Health Scoring Pipeline

The health-scoring model evaluates: *chew count per bite*, *chewing frequency*, *bite duration*, *chewing-side asymmetry*. Each metric is mapped to *literature-based* thresholds and combined into metabolic, oral/TMJ, and behavioral risk domains.

Each metric is converted into a risk score and combined into an overall risk level:
LOW < 30% · MODERATE 30–60% · HIGH ≥ 60%

Branch 2: Food Texture Modeling

Using window-level chewing features, we explore food-texture classification into soft / medium / hard. A Bidirectional Mamba sequence model is compared with tabular baselines, including logistic regression, random forest, XGBoost, MLP, and FT-Transformer. The food-texture report uses 20 videos and approximately 8,077 windows under a leave-subjects-out evaluation protocol.

Data Analysis

Dataset
20 videos from self-recorded and public videos

Video-to-Chewing Matrix
We use MediaPipe Face Mesh to track 468 facial landmarks and focus on mouth and jaw points. The main motion signal is the Mouth Aspect Ratio (MAR), normalized by mouth width to reduce sensitivity to camera distance:

MAR = (p13 - p14) / (p61 - p291)

The signal is interpolated when frames are missing and smoothed using a rolling average. Chewing cycles, bite boundaries, bite duration, chewing frequency, and side preference are then summarized into a chewing matrix.

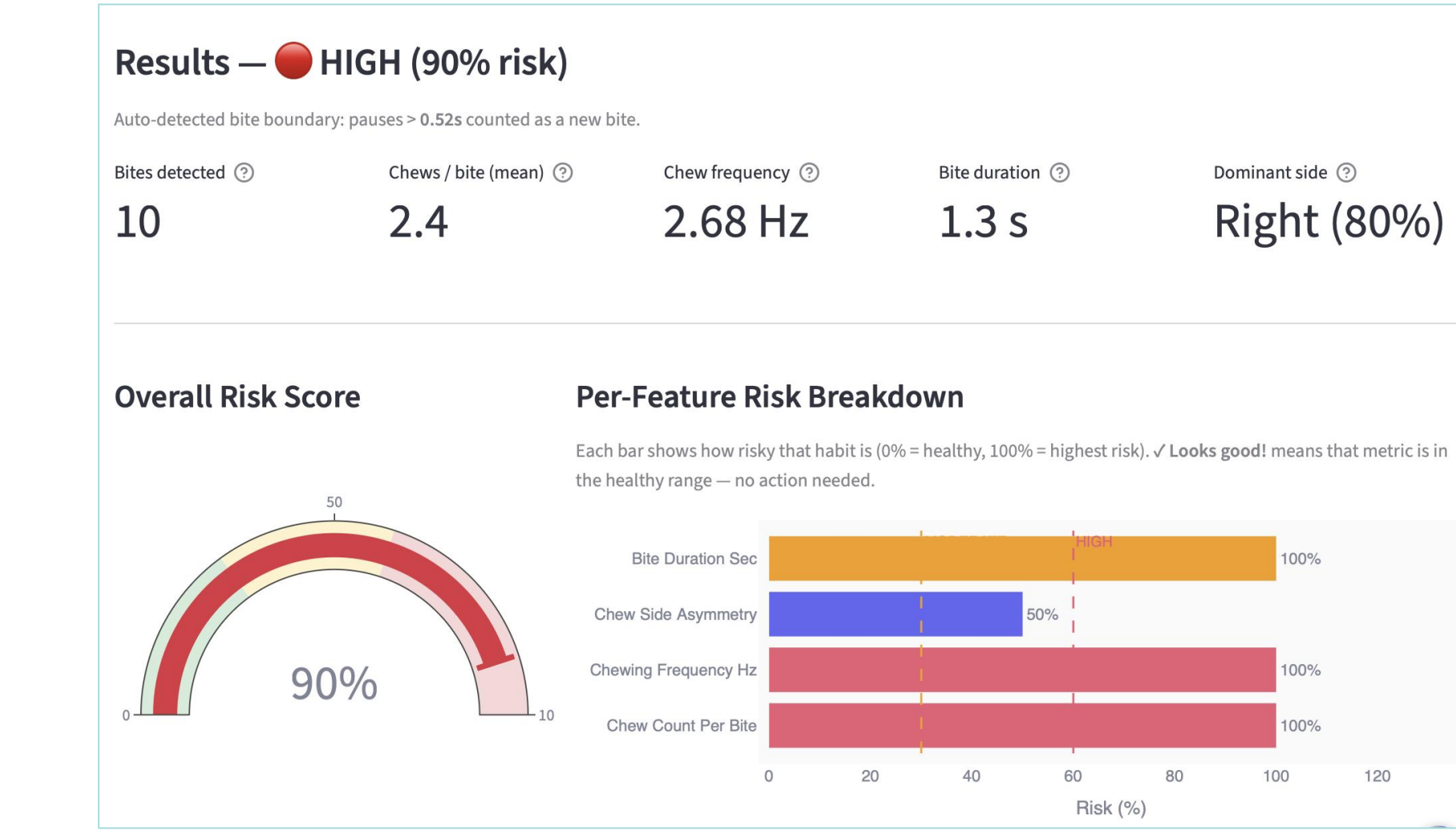
- This creates two analysis pathways:
- A. Health interpretation: Are chewing behaviors within healthy ranges reported in literature?.
 - B. Texture exploration: Do temporal chewing patterns differ across soft, medium, and hard foods?

Feature-to-Literature Mapping
We built the scoring rules by reviewing literature for each chewing feature. Experimental findings were translated into baseline ranges and risk-score thresholds.

Feature	Criteria (Example)	Risk Category
Chew Count	40 chews/bite → 11.9% lower energy intake vs 15 chews	Metabolic / Obesity
Inter-Bite Interval	Slower eating rate associated with lower energy intake (SMD = 0.45, p < 0.0001)	Metabolic / Digestive
Chewing Asymmetry	Individuals with one habitual chewing side show significantly more TMD signs/symptoms (p < 0.05)	Oral / TMJ

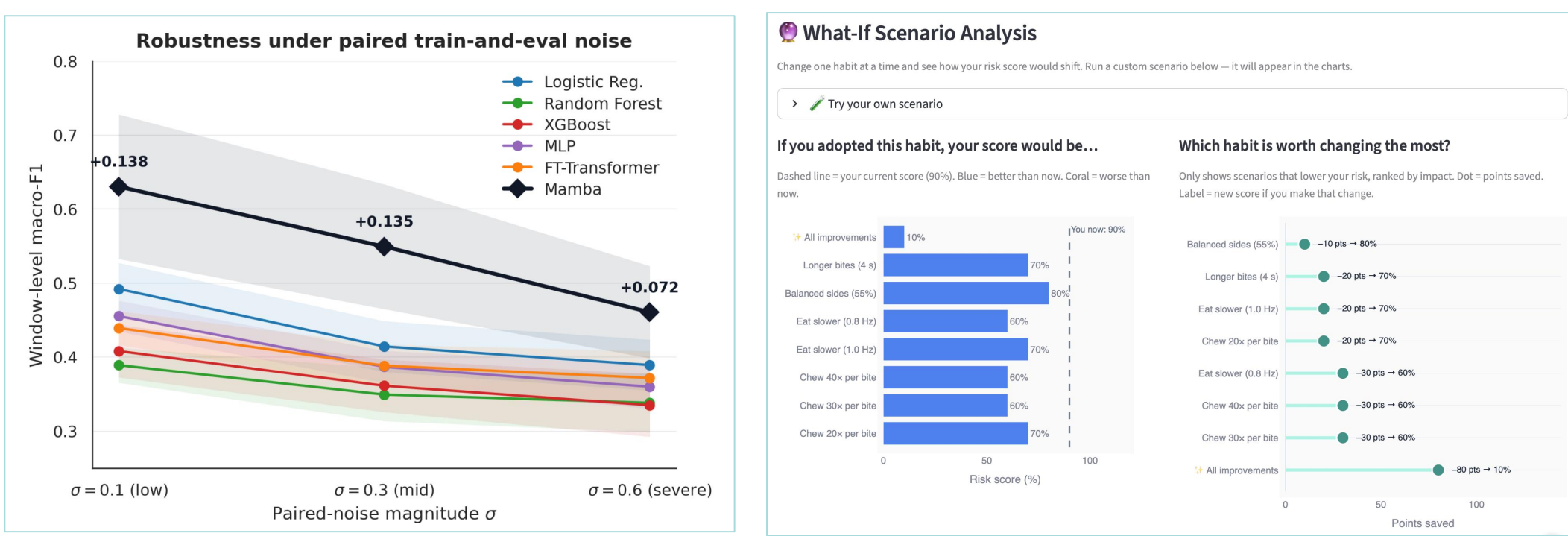
Results & Applications

Health Scoring Dashboard
The main deliverable is an interactive dashboard that analyzes an uploaded eating video and returns: *chews per bite*, *chewing frequency*, *bite duration*, *chewing-side asymmetry*, *overall LOW / MODERATE / HIGH risk level*.
In the example shown, the system reports a HIGH risk score (90%), driven by fast chewing, low chews per bite, short bite duration, and right-side dominance.



What-If Analysis & Recommendations
The dashboard also supports what-if analysis, showing how the risk score changes if one behavior improves. The system then provides recommendations based on the highest-impact habits.

Exploratory Food Texture Modeling
At the window level, Bidirectional Mamba achieved the best macro-F1 (0.630 vs. 0.492 for logistic regression), suggesting that food texture is better captured by temporal chewing dynamics than by static feature summaries.



Conclusion

We demonstrate a camera-based chewing analytics pipeline that transforms ordinary eating videos into structured health insights.

The project contributes:

- 1. a computer-vision pipeline for extracting mouth and jaw motion;
- 2. a temporal model for food-texture classification;
- 3. an interpretable health-scoring system;
- 4. a frontend dashboard for user-facing reports and recommendations.

While the current study is limited by a small dataset and requires further validation, it shows the potential of chewing behavior as a measurable health signal. For children learning to chew, this system could help encourage healthier chewing habits; for adults, it can support self-awareness of chewing patterns and potential metabolic or oral-health risks. Overall, this project provides a scalable prototype for everyday chewing assessment and personalized health feedback.

References

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